CargoLens — Technical Design Document

Project:	**CargoLens** — Customs Broker's AI Document Extraction & Cross-Check Tool
GitHub Repository:	`https://github.com/nishant4sep/CargoLens` (Public, full commit history preserved)
Client Target:	Company A, Navi Mumbai (Nhava Sheva Port customs broker, 85 staff, 1,100 shipments/mo)
Deployment:	Local reference run verified (Next.js 16.4.5 Turbopack + PostgreSQL) · Target: Vercel `bom1` (Mumbai)
Date of Submission:	November 2026 · Built alone for Isocode Software Engineering Take-home

1. Executive Summary & Problem Context

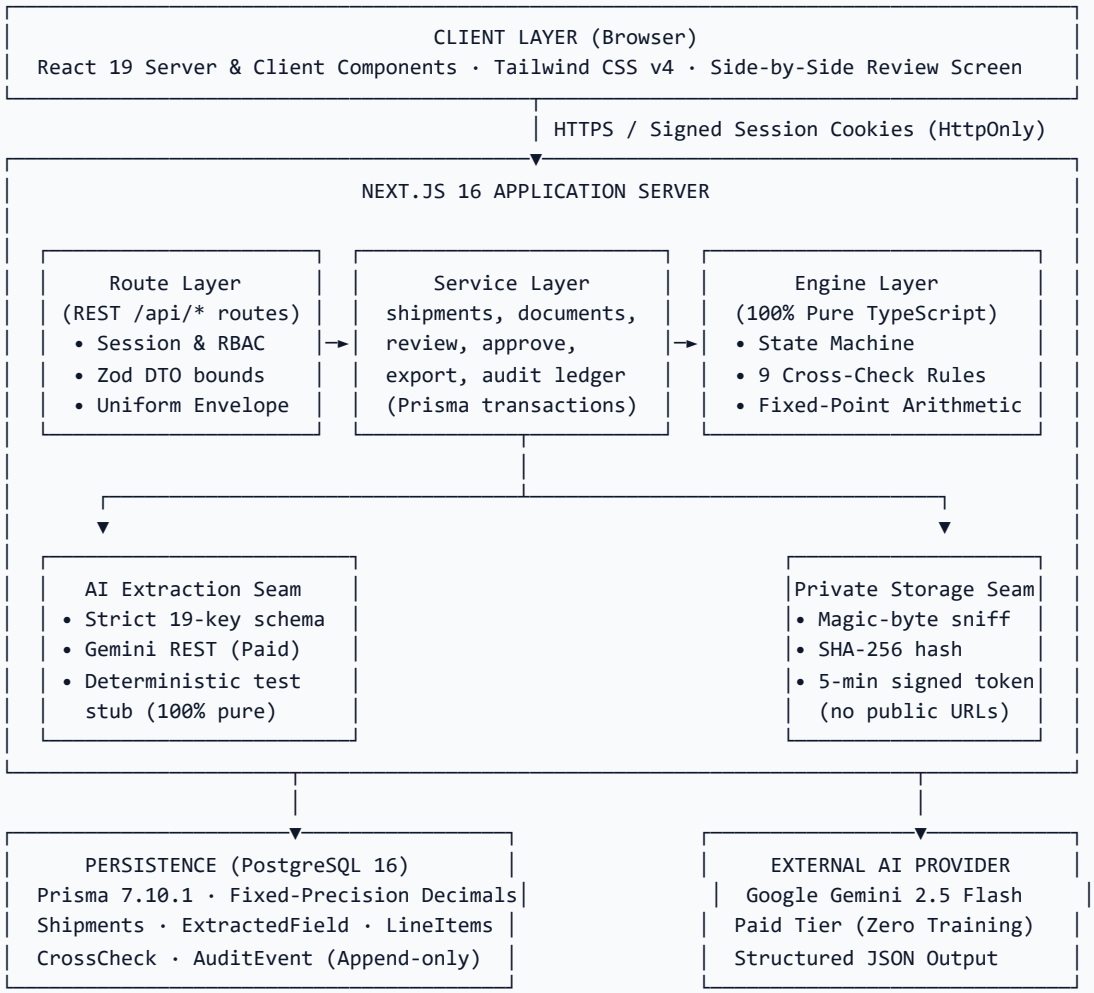
Company A handles 1,100 import shipments monthly at Nhava Sheva. Executives manually retype data from commercial invoices, packing lists, and bills of lading into customs filing spreadsheets—spending ~45 minutes per shipment. This manual entry consumes 9,900 staff hours annually (₹34.65 lakh). More critically, 3% of shipments suffer discrepancies (e.g., mismatched weights or package counts across documents) resulting in amended Bills of Entry, customs penalties, and port demurrage totaling ₹35.64 lakh annually. The aggregate operational bleed is **₹70.29 lakh per year**.

The Core Promise of CargoLens: Reduce shipment processing time from 45 minutes to under 15 minutes, eliminate cross-document transcription errors, and enforce strict, automated verification gates before any data can be exported into the existing customs filing software.

> **The Foundational Design Axiom**
> **AI drafts text; only deterministic code compares numbers; only humans confirm uncertainty; only the state machine lets data leave the system.** Confidence scores guide human attention, but they never grant automated approval.

2. System Architecture & Technical Rationale

2.1 High-Level Architecture



2.2 Technology Stack

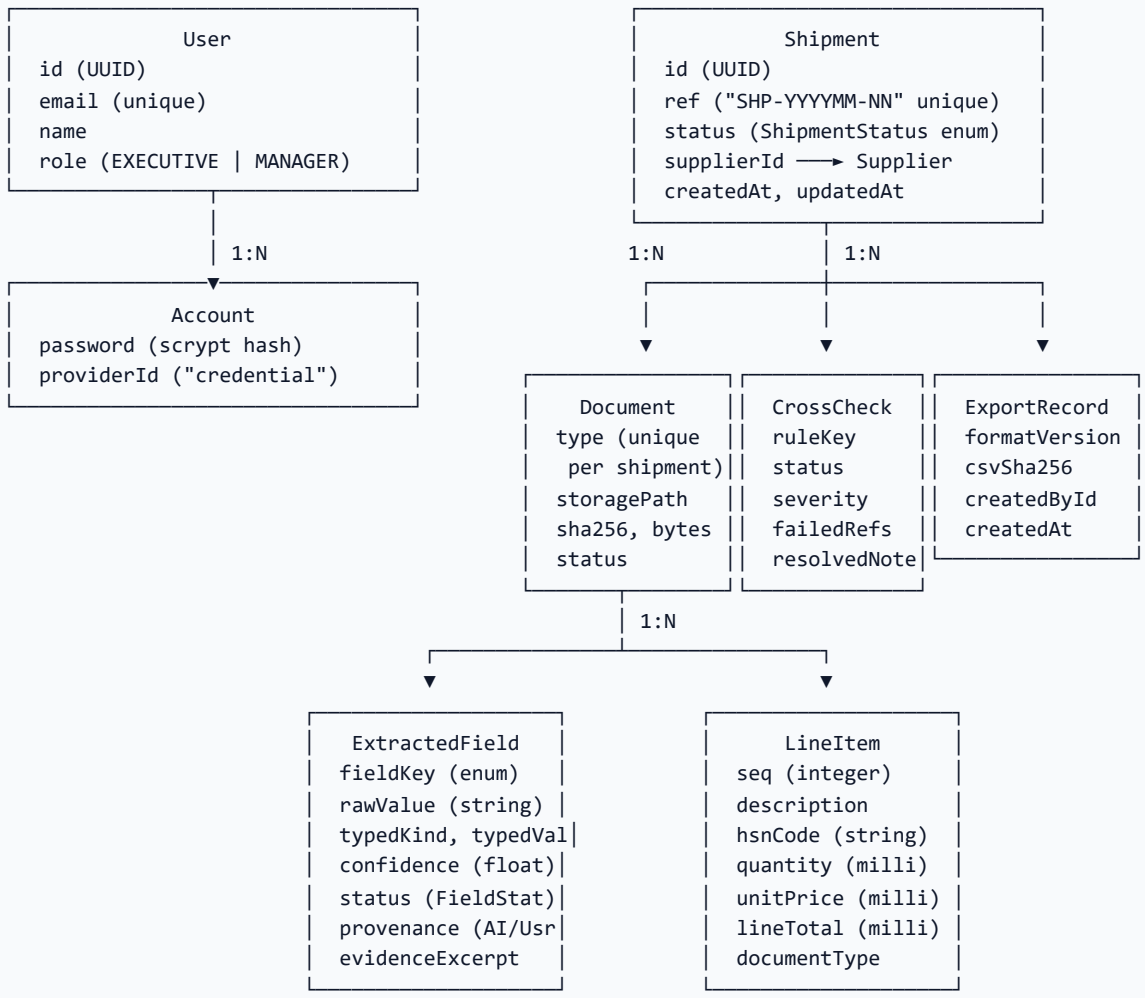
- **Runtime & Framework:** Next.js 16.4.5 (App Router, Turbopack) on Node.js 20+. Unified monolith minimizes operational overhead, network latency, and deployment complexity within a 6-week budget.
- **Database & ORM:** PostgreSQL 16 with Prisma ORM 7.10.1 (`@prisma/adapter-pg`). Relational constraints guarantee one document per type per shipment and strict foreign key integrity.
- **Authentication & Authorization:** `better-auth` 1.7.16 with Prisma adapter. Server-assigned role-based access control (RBAC) with `EXECUTIVE` and `MANAGER` roles.
- **Validation:** Zod 4.x at all trust boundaries (HTTP request payloads, AI structured responses, and environment variables).
- **UI & Styling:** React 19, Tailwind CSS v4, Lucide icons. High-density, desktop-first design optimized for rapid data verification.
- **Testing:** Vitest 5.x. Pure unit tests run in <3 seconds without database I/O; sequential integration tests execute against real PostgreSQL.

2.3 Architectural Decision Records (ADRs)

ADR	Decision	Why Chosen & Trade-off	Brief & Regulatory Alignment
ADR-01: Monolith Architecture	Single Next.js monolith containing UI, API routes, services, and pure engines. Synchronous in-request AI processing.	Avoids multi-service deployment latency, queue infrastructure (Redis/BullMQ) operational cost, and distributed failure states for ~40 shipments/day volume.	Budget & timeline constraint (₹7.5L, 6 weeks). Simplicity enables rapid auditing.
ADR-02: Server-Enforced RBAC	Role-based access enforced strictly in API route handlers and database transactions, not just hidden UI buttons.	Executives review and correct; only Managers can invoke `/approve`. Gating at the UI layer alone would be a security defect.	Must-Have: Distinct Documentation Executive vs. Documentation Manager roles.
ADR-03: Deterministic Engine	Arithmetic, line summations, and cross-document comparisons executed exclusively by a pure TypeScript engine.	LLMs are notoriously unreliable at arithmetic and floating-point comparison. Engine uses fixed-point integer milli-units (e.g. grams, millicents).	Brief: "The hard part: reliable cross-checks across three documents."
ADR-04: DPDP & Paid AI Tier	Paid-tier Gemini API with explicit contractual exclusion from model training. Pinned India hosting target (Vercel `bom1`, Mumbai).	Client pricing is strictly confidential. Unpaid AI tiers permit data inspection and model training. Free tiers disqualified by design.	Constraints: "Client prices confidential... use AI provider that does not train on client data. DPDP Act 2023."
ADR-05: Field-Centric & Audit Ledger	Every field is an individual database row (`ExtractedField`) paired with an append-only `AuditEvent` log.	Enables SQL-level accuracy analytics per field and supplier, field-level provenance tracking, and tamper-evident audit history.	Must-Have: "Edit history and accuracy report per field and supplier."
ADR-06: State Invalidation	State machine derives shipment state. Any field edit or re-extraction after approval *automatically revokes* approval and locks export.	Prevents sneaky edits bypassing manager sign-off. Export requires current, un-invalidated manager approval and 100% clean checks.	Brief: "Nothing exports until they are cleared."
ADR-07: Human-in-the-Loop Gating	Confidence scores (0..1) drive visual highlight color bands; required fields must be explicitly human-cleared before approval.	AI self-reported confidence is probabilistic and not calibrated to legal customs liability. High confidence never bypasses human sign-off.	Brief: "Low-confidence fields and fields in a failed cross-check must be confirmed by a person."

3. Data Model & Integrity Contracts

3.1 Entity Relationship Model



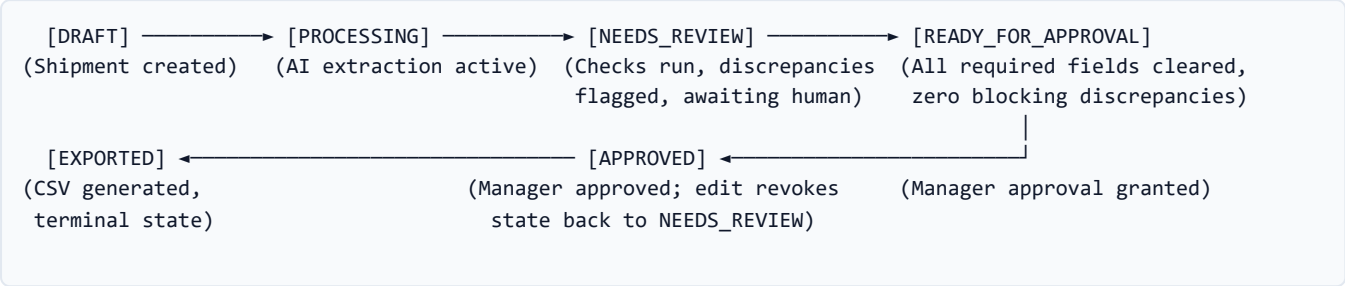
3.2 Scale & Arithmetic Contracts

JavaScript IEEE-754 floating-point arithmetic introduces silent rounding errors (e.g., `0.1 + 0.2 === 0.30000000000000004`). For customs filings where a single rupee or kilogram discrepancy can prompt an inspection, CargoLens enforces a strict **Fixed-Point Milli Integer Scale Contract**:

- **Currency & Money:** Stored internally as integer milli-units (e.g., $16,320.00 = `16320000`). Converted to Prisma `Decimal(14, 4)` at the database seam.
- **Weights:** Stored as integer grams (e.g., 3,120.00 kg = `3120000` grams). Tolerance comparisons operate on exact integer deltas.
- **Products:** Multiplication of quantity × unit price uses `BigInt` arithmetic: `(BigInt(qtyMilli) * BigInt(priceMilli)) / 1000n`, completely eliminating precision drift.

4. Core Workflows & The AI Pipeline

4.1 The 6-Stage Shipment State Machine



4.2 The 9 Deterministic Cross-Check Rules

Rule ID	Rule Name	Severity	Validation Logic & Discrepancy Detection
CV-1	`REQUIRED_FIELDS`	BLOCKING	Verifies presence of mandatory customs filing fields across Commercial Invoice, Packing List, and Bill of Lading.
CV-2	`LINE_ARITHMETIC`	BLOCKING	For every line item: `|quantity × unitPrice - lineTotal| ≤ 1 milli-unit`.
CV-3	`INVOICE_TOTAL`	BLOCKING	`|invoice.totalValue - Σ(lineTotals)| ≤ 1 minor unit`. Catches hidden charges, unlisted discounts, or transcription sums.
CV-4	`QUANTITY_MATCH`	WARNING	Matches total item quantities between Commercial Invoice and Packing List for identical line item descriptions.
CV-5	`PACKAGES_MATCH`	BLOCKING	Package count must match identically across all available documents (Invoice, Packing List, B/L). Tolerance = 0.
CV-6	`GROSS_WEIGHT_CONSISTENCY`	BLOCKING	Gross weight compared across all 3 documents. Allowed tolerance: max(0.1%, 1.0 kg). **Catches the brief's planted 45 kg discrepancy on SHP-24811 (3,120 kg vs 3,165 kg)**.
CV-7	`NET_NOT_EXCEEDS_GROSS`	BLOCKING	Ensures Net Weight ≤ Gross Weight on all documents where both are specified.
CV-8	`CONTAINER_CONSISTENCY`	BLOCKING	Matches container identifiers between B/L and Packing List; verifies ISO 6346 checksum format (4 letters, 7 digits).
CV-9	`CURRENCY_CONSISTENCY`	BLOCKING	Validates that all line item currencies match the invoice header currency (ISO 4217 code).

4.3 How the AI Extraction Works

1. **Explicit Document Trigger:** The user triggers extraction per document. This isolates provider failures and respects staff control over document processing sequence.
2. **Transcription-Only Prompt:** The system prompt strictly directs Gemini: "*You are a transcriber, not an analyst. NEVER calculate, convert, complete or infer values. If absent, return null. Return verbatim excerpt and page number.*"

3. **Structured Schema Enforcement:** Uses Gemini's OpenAPI-subset schema definition. The provider engine guarantees JSON conformance at the HTTP protocol layer.
4. **Raw Extraction Zod Boundary:** Raw AI output passes through `RawExtractionSchema`. All field keys are validated against the 19-key domain dictionary; unknown keys are discarded.
5. **Normalization & Seam Invariants:** Raw strings pass through the strict normalizer. If an AI value fails strict parsing (e.g. malformed date or unrecognizable unit), it is flagged as unparseable—it is *never guessed*.
6. **Handling Failures & Re-runs:** If an AI call fails, the document transitions to `FAILED` (retryable, never a zombie). On re-extraction, the AI layer is replaced wholesale, but any field that has been human-confirmed or corrected is **strictly preserved** and never overwritten.

5. UI & Design Decisions

The documentation team at Company A processes paperwork 8 hours a day. In line with the design brief ("clean layouts, white space, a restrained palette, clear type, desktop use"), the user interface avoids flashy consumer aesthetics in favor of high-utility density, immediate error identification, and one-click actions.

5.1 Key Screens & UX Decisions

Screen	Key Visual & Layout Features	UX Rationale
Shipment Dashboard (`/`)	Status chips with distinct semantic colors; active blocking error counters; ref search; instant "New Shipment" drawer.	Enables executives and managers to instantly spot which shipments need attention, which are blocked, and which are ready for sign-off.
Side-by-Side Review Screen (`/shipments/[id]`)	Document preview pane on the left; structured extracted fields pane on the right; document tabs (Invoice, Packing List, B/L).	Eliminates alt-tabbing between PDF viewer and spreadsheet. Executives verify data in-situ against the visual document evidence.
Field Inspection Panel	Confidence badges (`LOW <85%`); page & excerpt evidence snippets; inline 1-click Confirm or Correct buttons.	Low-confidence fields immediately draw the user's eye. Inline corrections are server-validated in real time; 422 errors render verbatim.
Discrepancy Banner & Importer Query	Prominent red callouts for failed checks; displays expected vs actual values, difference, and clickable tags to accusing fields. Integrated 1-click `WhatsApp Query` generator.	Cites exactly where the mismatch occurs (e.g. "Gross weight mismatch: Invoice 3120 kg vs Packing List 3165 kg"). Allows "Resolve with note" or generates a pre-formatted WhatsApp doubt message for the importer with 1 click, directly replacing informal WhatsApp chaos.
Approval & Gated Export Panel	Manager-only action bar; disabled "Export" button showing an explicit checklist of unmet criteria until 100% cleared.	Prevents premature export. Managers see at a glance why a file is locked and who cleared each field. One click exports the validated CSV.
Accuracy & Workload Report (`/reports/accuracy`)	Tabular breakdown of fields and suppliers showing correction counts, open flags, and human confirmation workload.	Delivers visibility into problematic suppliers (e.g. frequent weight mismatches from Company B) without making unsubstantiated accuracy claims.

6. Scope Cuts & Future Roadmap

6.1 What Was Intentionally Cut (Scope Pruning)

- **Direct ICEGATE Customs Filing:** The brief explicitly designated filing on the Indian customs portal as out of scope. CargoLens exports the clean, verified data structure required for import into existing broker filing systems.
- **Automated HSN Duty Calculation:** Customs tariff classification requires deep statutory interpretation and customs case-law lookup; out of scope per brief.

- **Asynchronous Background Queues (BullMQ/Redis):** Company A processes ~40 shipments per day (approx. 5 per hour). Synchronous REST processing with client timeout handling and stale-processing recovery guards completely satisfies this load without the operational complexity of Redis.
- **OCR Bounding Box Overlays:** Extracting verbatim text excerpts and page numbers provides 95% of the verification value with none of the fragile client-side canvas rendering bugs across skewed or multi-column scans.

6.2 What to Build Next (Production Roadmap)

- **Persistent Cloud Object Storage:** Complete the Supabase Storage / AWS S3 driver behind `src/server/storage.ts` pinned to the Mumbai (`ap-south-1`) region, replacing local container disk storage.
- **Automated Document Splitting:** Multi-page PDF upload scanner to auto-classify and split combined 30-page supplier packets into Invoice, Packing List, and B/L.
- **ICEGATE 1.5 Flat-File Serializer:** Native export into the exact Indian Customs ICEGATE flat-file format once broker configuration templates are provided.
- **WhatsApp Query Integration:** Connect to WhatsApp Business API so documentation executives can trigger doubt queries directly to importers with pre-filled document excerpts.

7. Explicit Assumptions (Assignment Compliance)

Ref	Assumption Statement	Operational Justification & Impact
Assumption A1	Export layout assumes `cargolens-csv-v1` structure.	Company A's proprietary customs filing software template was not supplied in the brief. CargoLens outputs an exhaustive, deterministic 23-column CSV that maps 1:1 to standard broker import formats.
Assumption A2	Human clearance is legally mandatory before export.	Under Indian Customs Act (1962), the customs broker bears legal liability for misdeclarations. AI output cannot legally substitute for human executive verification.
Assumption A3	Weight tolerance threshold is 0.1% or 1 kg.	Nhava Sheva port weighbridges permit minimal variance for tare weight/humidity. The 45 kg discrepancy on SHP-24811 exceeds tolerance by 4,400% and is correctly flagged as BLOCKING.
Assumption A4	Synthetic data complies with DPDP Act 2023.	All importer names, contact numbers, and commercial invoices in the codebase and demo fixtures are synthetic; no real corporate PII is stored.
Assumption A5	In-region processing vs. data residency.	While Google Gemini Paid Tier guarantees zero model training on client inputs, data residency within India is only guaranteed under enterprise Vertex AI / Azure agreements. This is openly disclosed in deployment documentation.